Information for Candidates

Switch-Mode Power Supplies

Voltage ratio equations for buck SMPS $\frac{V_o}{V_i} = \delta$ or $\frac{V_O}{V_I} = \frac{1}{1 + \frac{2fLI_O}{V_I\delta^2}}$

Voltage ratio equations for boost SMPS $\frac{V_o}{V_i} = \frac{1}{1-\delta}$ or $\frac{V_o}{V_i} = \frac{1}{1 \quad \frac{V_i^{\ 2}}{2fLI_i}}$

Three-Phase Systems

Line Voltages and Current

Star $V_L = \sqrt{3}V_P \quad I_L = I_P$ Delta $V_L = V_P \quad I_L = \sqrt{3}I_P$

$P_3 = 3V_PI_P\cos(\ )$

Power $= \sqrt{3}V_LI_L\cos(\ )$

Induction Machines

$S = \frac{E}{P} \quad s = \frac{S \quad R}{S} \quad T_{em} = \frac{3I_R^{'2}R_R^{'}}{\omega_R}\left(\frac{1-s}{s}\right)$

Photovoltaic Systems

$I_{PV} = I_{Ph} - I_{AK} - I_{Sh} \quad I_{AK} = I_0\left[e^{\frac{V_{PV}+I_{PV}R_S}{K_I v_T}} - 1\right]$

Power Flow in Lines and Cables

Cable Parameters

$R'_{LF} = \frac{1}{\ _C r_C^2} + \frac{1}{\ _C 2\ r_O t_O} \quad L' = \frac{\ _0}{2}\ln\ ^{r_O}\!/_{r_C} \quad C' = \frac{2\ _{0\ RI}}{\ln\ ^{r_O}\!/_{r_C}} \quad G' = \frac{2\ _I}{\ln\ ^{r_O}\!/_{r_C}}$

OHL Parameters (approximate form)

$R'_{LF} = \frac{2}{\sigma_C\pi r_C^2} \quad L' = \frac{\mu_0}{\pi}\ln\left(^{d}\!/_{r_C}\right) \quad C' = \frac{\pi\varepsilon_0\varepsilon_{RI}}{\ln\left(^{d}\!/_{r_C}\right)} \quad G' = \frac{\pi\sigma_I}{\ln\left(^{d}\!/_{r_C}\right)}$

Power Flow (full form)

$P_S = \frac{V_S^{\ 2}}{Z_{SR}}\cos(\ ) \quad \frac{V_SV_R}{Z_{SR}}\cos(\ +\ )$

$Q_S = \frac{V_S^{\ 2}}{Z_{SR}}\sin(\ ) \quad \frac{V_SV_R}{Z_{SR}}\sin(\ +\ )$

Voltage Drop (approximate form)

$\Delta V = |V_S| - |V_R| \approx \frac{RP_S + XQ_S}{|V_S|}$

Answers

[Overall candidates performed well on this paper and achieved an average mark of 64.1%. The average marks on each question were: Q1 24.5/40 (61%), Q2 18.7/30 (62%) and Q3 21.1/30 (70%). Q1 was on a wide range of basic concepts in power engineering where perhaps it is the breadth rather than the depth that makes it a challenge. Q2 was on a topic that most students find approachable but was pitched to be the difficult question on this paper that would stretch candidates. In the light of the difficulty, the candidates performed very well. Q3 was on a familiar topic and similar in format to a question on the problem sheet so the mark was somewhat higher.]

1. This question covers several topics and all parts should be attempted.

a) Consider a national scale electricity system based on AC.

i) Describe the principal difference between the transmission and distribution networks and explain why transmission networks use higher voltages than distribution networks. [6]

Transmission networks exist to effect bulk transfer of power from geographic regions with more generation than load to regions with more load than generation. They are long distance networks which are built to achieve high availability by having duplicate circuits on all routes and having a high degree of meshing between the nodes. Distribution networks a relatively short distance networks for distributing power within a load centre. They use a variety of voltage levels determined by the volume of load being served. There are double circuit routes at the higher voltages and single circuit routes at the lower voltages but little or no meshing.

[4 marks for mentioning degree of redundancy and availability, difference in function and difference in geographic scale]

The choice of operating voltage is based on achieving the lowest total cost considering operating costs and capital costs. Higher voltages reduce power losses and hence operating costs but have higher capital costs. For long distance routes with relatively few nodes and substations, high voltage achieves a large advantage in power loss for reasonable capital cost. Where distances are short, and connections are many, the lower voltages achieve a better balance between the costs.

[2 marks]

[The basic purpose of the transmission network (bulk transfer from generation to load regions) and distribution network (delivering power within a load centre) were often correctly cited, but often the trade-off between capital cost of high voltage equipment and power losses over distances was inadequately explained. Transmission networks operate over long distances, have few nodes which serve many customers. Therefore, the solution consists of HV (higher capital cost but fewer nodes and lower power losses thanks to lower I^2R over longer distances) and highly meshed grids (due to high cost of outage); and vice versa with distribution networks.

Furthermore, a common mistake consisted of stating that the voltage magnitude is reduced in distribution networks due to safety concerns. Even reduced, 33kV down to 240V magnitudes remain potentially lethal.]

ii) Most countries of the world operate their electricity networks at a nominal frequency of 50 or 60 Hz, however, in practice the frequency is not strictly constant. Explain why the frequency varies a little around its nominal value. [4]

The frequency is determined by the rotational speed of the generators of the system. These generators are locked-in to each other and run in synchronism but their combined speed will still increase if there is a net torque applied to their combine inertia. Thus if there is a mismatch between the power (torque) applied to the generator shafts by the prime movers and the power extracted (reaction torque) by the electrical loads then the speed will change. If there is an excess of electrical load, then the "missing"

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power is extracted from the kinetic energy of the system until such time as extra power is applied from the prime movers.

[It was clear that many candidates did not carefully read the question before answering. In this case, a very large number of answers focused on the choice of the nominal generation frequency of 50 Hz or 60 Hz rather than why the actual grid frequency may fluctuate ±0.1-1 Hz around these nominal values.

Furthermore, the answer lies in the mechanical aspect of the electrical grid, with all the generators synchronised together as one big inertial machine, which slows down if the electrical load is larger than the mechanical input power; and vice versa. Some of those who did attempt to answer the right question grasped unrelated factors such as the size of the transformers or the slip ratio of induction machines]

b) Consider power transfer in AC networks.

i) A single-phase cable operated at 50 Hz and 120 kV has a capacitance per unit length of 0.25 μF/km. Calculate the capacitive charging current for a 75 km length of cable. [2]

Use standard formula and assume that approximately the same voltage is found along the length of the cable.

$$I_C = \frac{V}{X_C} = V \times 2\pi \times f \times d \times C' = 120k \times 2\pi \times 50 \times 75 \times 0.25\mu = 706.9 \quad A$$

[Most common mistake was to not properly include the factor of 2π (radian)]

ii) A single-phase overhead line has a series impedance which is purely reactive with a value of 18 Ω and is operated at a voltage of 400 kV. Calculate the phase angle that would exist between sending and receiving end voltages when transferring 1,000 MW. [4]

Use standard formula for power transfer across inductive impedance.

$$P = \frac{V^2}{X}\sin(\delta)$$

$$\delta = \sin^{-1}\left(\frac{XP}{V^2}\right) = \sin^{-1}\left(\frac{18\times10^9}{\left(400\times10^3\right)^2}\right) = 6.46°$$

[The correct formula uses sin(δ) and NOT cos(δ) which leads to an erroneous answers of 84º.

Furthermore, the question states that the line has a purely reactive series impedance, NOT that the total impedance is reactive. Therefore, the phase angle cannot be 90º.]

iii) A three-phase load is composed of impedances of 10 + j2 Ω connected in star. It is connected to a supply with a line voltage of 400 V. Calculate the real and reactive power drawn. [4]

[2 marks for method, 2 for accurate answers]

First find phase voltage and phase current.

Answers

$$V_P = \frac{V_L}{\sqrt{3}} = 230.9 \quad V$$

$$I_P = \frac{V_P}{Z} = \frac{230.9}{10 + j2} = 22.65\angle\text{-}11.3°$$

$$P = 3V_P I_P \cos(\theta) = 3 \times 230.9 \times 22.65 \times \cos(-11.3°) = 15.39 \quad kW$$

$$Q = 3V_P I_P \sin(\theta) = 3 \times 230.9 \times 22.65 \times \sin(-11.3°) = 3.08 \quad kVAr$$

[Most candidates answered this correctly. The only notable mistakes were (i) computing the power for only one phase (5kW/1kVAr) and (ii) misuse of the calculator including degree/radian mode confusion.]

c) Compare buck and boost switch mode power supplies (SMPSs) through the following steps.

i) For a fixed input voltage, sketch the shape of the output voltage as a function of the duty-cycle of the switch (assuming continuous conduction) for each SMPS. [3]

Answer in form graph expected but important aspects of shape are as follows. For Buck case the graph is linear starting at the origin and reaching V_I at δ=1. For Boost case, graph is non-linear, starts at V_I pass through 2 V_I at δ=0.5 and tends to infinity as δ approaches 1.

[Some candidates did not read the question carefully and answered instead with the switching pattern of SMPSs.

Furthermore, although a sketch is asked for, a key features need to be accurate and annotation is helpful in this regard. The buck converter voltage ratio goes from 0 to 1 linearly with the duty cycle (also only from 0 to 1). The boost converter voltage ratio starts at 1 for a 0 duty cycle, then 2 for a duty cycle of 0.5 and approaches infinity nearing a duty cycle of 1].

ii) Sketch the shape of the currents as a function of time for the currents that flow in the inductor, diode and capacitor for each type of SMPS. [5]

Answer in form graph expected but important aspects of shape are as follows. Inductor current and diode current are similar for Buck and Boost, inductor is DC with triangular ripple, diode is trapezoid with downward slope matching part of inductor current [2 marks]. Capacitor current is different in the two cases, for Buck it is the triangular ripple portion of the inductor current [1 mark], for boost it is the diode current shift down by the output current such that it has an average value of zero [2 marks].

[As in part (i) key features need to be accurate and annotation is helpful in order to demonstrate the required degree of understanding of the concept. The sketches of the waveforms for each SMPS type should be aligned in time (perhaps drawn on the same axis or with some labelling of he time axis to show key timing points.

The most common mistakes were over the current waveform of the capacitor in the boost circuit. The overall shape is exactly the same as the diode (NO inversion of the slope direction or timing) and shifted vertically by the output current magnitude (which should be by nature smaller than the input current).]

iii) For similar magnitudes of output current and the same choice of capacitor, compare the amplitudes of output voltage ripple. [2]

The trapezoidal pulse of capacitor current in boost is about the amplitude of the input current (which is large than the output current for step up operation). The triangular ripple of the Buck is smaller than the output current. Thus the voltage drop across the ESR and the C is larger in the Boost case.

Answers

[Generally answers here were good. Some lacked a (short) explanation on why the boost output voltage ripple is bigger than for the buck.]

d) Consider a typical photovoltaic panel.

i) Describe the operating principle of a photovoltaic cell and the role of the band-gap energy in determining the efficiency of the cell. [5]

The cell uses the photoelectric effect in which a photon of sufficient energy can ionise an atom in the lattice of a semiconductor and generate a hole-electron pair. That pair of charge carries are swept out of the depletion region and appear as external current. [2 marks]

To generate the hole-electron pair, the photon must have at least the energy of the semiconductor band-gap. Photons with less energy are not absorbed and this is an aspect of the inefficiency of the cell. Photons with more energy than the band-gap create a pair but only of the energy of gap is gained as useful energy, the remainder is lost by the electron to the lattice as heat which is also an inefficiency. [3 marks]

[The fact that ONLY the energy equal to the bandgap is converted to electrical form (anything below is unconverted, any extra above is lost as heat) was sometimes misreported, leading to erroneous claims that higher bandgap PV cells are more efficient.]

ii) A PV cell was tested at 25˚C. It produced a short-circuit current of 0.8 A under an irradiance of 250 W/m^2. The reverse saturation current of the cell was measured as 1×10^{-10} A. You may assume that the cell has a very small series resistance, a very large shunt resistance and an ideality factor of 1. Boltzmann's constant is $k = 1.380\times10^{-23}$ J/K and the electronic charge is $q = 1.602\times10^{-19}$ C. Calculate the open-circuit voltage at irradiances of 250 and 1,000 W/m^2. [5]

In the open-circuit case all of the photoelectric current flows through the diode-like junction governed by the diode equation:

$$I_{AK} = I_0\left[e^{\frac{q\,V_{AK}}{K_I\,kT}} - 1\right]$$

Setting K=1 and re-arranging give the equation for the voltage in terms of the current.

$$V_{AK} = \frac{kT}{q}\ln\left[\frac{I_{AK}}{I_0} + 1\right]$$

At 250 W/m^2, the current is 0.8 A and so:

$$V_{AK} = \frac{(25+273)\times1.380\times10^{-23}}{1.602\times10^{-19}}\,ln\left[\frac{0.8}{10^{-10}} + 1\right] = 0.5854$$

[3 marks]

At 1,000 W/m^2, the current is four times greater, i.e. 3.2 A and so:

$$V_{AK} = \frac{1.380\times10^{-23}}{1.602\times10^{-19}}\,ln\left[\frac{3.2}{10^{-10}} + 1\right] = 0.6209$$

[2 marks]

Answers

[Usually after remembering that $V_T = \frac{kT}{q}$, then most of the candidates applied the right formula and obtained the correct answer.

One occasional mistake was to multiply the open circuit voltage by 4, instead of the current, in the 1000 W/m^2 case]

Answers

2.
A **buck** switch-mode power supply, SMPS, is to be designed to provide a 5 V output voltage from a 24 V input. The output current will be in the range 0.1 A to 2.0 A and it is intended that the SMPS operates in continuous conduction mode across this range. It has been decided to design for a switching frequency of 150 kHz.

A suitable MOSFET has been identified for the SMPS with the following properties:

On resistance $R_{DS(on)} = 0.25\ \Omega$
Turn-on loss $E_{on} = 13\ \mu J$ when switched on at 2.0 A
Turn-off loss $E_{off} = 7\ \mu J$ when switched off at 2.0 A

The diode to be used has an on-state voltage of 1.0 V.

i) Calculate the duty-cycle at which the switch will operate. [2]

Use standard formula

$$\delta = \frac{V_O}{V_I} = \frac{5}{24} = 0.2083$$

ii) Calculate the power losses in the semiconductors and the efficiency (ignoring other losses) when operating at maximum output power. [8]

Mosfet conduction loss [2 marks]

$$P_{Cond} = I_O^2 R_{DS} \delta = 2.0^2 \times 0.25 \times 0.2083 = 0.2083 \quad W$$

Mosfet switching loss [2 marks]

$$P_{Sw} = \left(E_{on} + E_{Off}\right) f = \left(13\mu + 7\mu\right) \times 150k = 3.0 \quad W$$

Diode conduction loss [2 marks]

$$P_{Diode} = I_O V_{AK} \left(1 - \delta\right) = 2.0 \times 1.0 \times \left(1 - 0.2083\right) = 1.58 \quad W$$

Total loss and efficiency [2 marks]

$$P_{Loss} = P_{Cond} + P_{Sw} + P_{Diode} = 0.2083 + 3.0 + 1.58 = 4.79 \quad W$$

$$\eta = \frac{P_O}{P_O + P_{Loss}} = \frac{10}{14.79} = 68.0\%$$

[Most common mistakes were oversights: neglecting to calculate the power loss in the diode or neglecting to calculate the efficiency. The most common error in calculation was to use an inappropriate formulation of the efficiency. Working with the output power and adding the losses to gain the input power is correct because the operating point has been specified in terms of output voltage and output current. Working with the input power, that is, $\eta = \frac{P_I - P_{Loss}}{P_I}$ because the input power current is unknown and $I_I = I_O \frac{V_O}{V_I}$ can't be used because it assumes 100% efficiency.

Answers

$\eta = \frac{P_O - P_{Loss}}{P_O}$ is simply incorrect.

A further occasional error was to use the input current in place of the output current when calculating the conduction losses. The output current is equivalent to the average inductor current and it is this current that is commutated between the Mosfet and diode.]

iii) Choose an inductor to ensure continuous conduction. [4]

Critical conduction condition is when output current is at half the inductor ripple current. Choose inductor to set the ripple current accordingly. This will be the minimum inductor value required. [2 marks for method; 2 for accurate answer].

$$\Delta I_L = 2I_O^{Critical} = 2I_O^{Min} = 2 \times 0.1 = 0.2 \quad A$$

$$\Delta I_L = \frac{V_I - V_O}{L} \times \frac{\delta}{f}$$

$$L = \frac{V_I - V_O}{\Delta I_L} \times \frac{\delta}{f} = \frac{(24-5) \times 0.2083}{0.2 \times 150 \times 10^3} = 132 \quad \mu H$$

[Generally this part was answered well but occasional mistakes included apply the factor of 2 in the first line in the wrong sense and using the largest, not the smallest, output current as the critical case. A careless slip that happened several times was to interpret $L = 1.32 \times 10^{-4}$ as 13.2 mH not 0.132 mH]

iv) Choose a capacitor to ensure that the output voltage ripple is less than 40 mV. You may work on the basis that 80% of the ripple will arise from the effective series resistance of the capacitor (ESR) and 20% from the capacitance. [6]

The inductor current ripple goes on to flow through the capacitor and cause the voltage ripple. Standard formulae apply.

[2 marks for method; 2 for each accurate answer].

$$\Delta v_{ESR} = 0.8 \times \Delta v_O = 0.032$$

$$R_{ESR} = \frac{\Delta v_{ESR}}{\Delta I_L} = \frac{0.032}{0.2} = 0.16 \quad \Omega$$

$$\Delta v_C = 0.2 \times \Delta v_O = 0.008$$

$$C = \frac{\Delta I}{8 f \Delta v_C} = \frac{0.2}{8 \times 150 \times 10^3 \times 0.008} = 20.8 \quad \mu F$$

[It was surprisingly common to neglect to calculate the required ESR of the capacitor. The question asked candidates to choose a capacitor, a physical component, which alongside choosing its capacitance requires other key parameters of the component to be specified such as its ESR. One could argue that you should also state that it needs a voltage rating of at least 5 V. A mistake made by several candidates was to misinterpret the split of the ripple voltage between ESR and C and state $\Delta v_O = 0.8\Delta v_{ESR} + 0.2\Delta v_C$. Not only is this incorrect but leads to single equation with two unknowns.]

Answers

v) Calculate the switching frequency that would be needed to improve the efficiency to 80%. [6]

Reducing the switching frequency will create a proportionate decrease in power loss due to switching. Begin by calculating the allowed total power loss for the efficiency given and then find allowed switching loss.

[4 marks for method; 2 for accurate answer].

$$\eta = \frac{P_O}{P_O + P_{Loss}}$$

$$P_{Loss} = \left(\frac{1}{\eta} - 1\right) P_O = \left(\frac{1}{0.8} - 1\right) \times 2 \times 5 = 2.5 \quad W$$

$$P_{Sw} = P_{Loss} - P_{Cond} - P_{Diode} = 2.5 - 0.2083 - 1.58 = 0.7083 \quad W$$

$$f = \frac{P_{Sw}}{E_{On} + E_{Off}} = \frac{0.7083}{(13\mu + 7\mu)} = 35.4 \quad kHz$$

[Candidates who answered part (ii) correctly generally managed to answer this correctly also.]

vi) What design changes would be needed (trends rather than numerical values are sufficient) as a consequence of the change of frequency determined in part (v)? [4]

The lower overall power loss from the lower switching loss will allow a smaller heat-sink to be used.

The lower frequency would cause a larger ripple current and so a larger range of output current causing discontinuous operation. If the constraint of part (iii) is retained, then a larger inductor (in inverse proportion to frequency reduction) will be required. If this is done, then the ESR specification for the capacitor is unchanged but a larger capacitance value is required. This may not change the capacitor choice if it was the ESR that was the dominant factor.

Answers

3.

A 2 pole-pair, 3-phase induction machine is star-connected and supplied at 50 Hz. The equivalent circuit of the machine is shown in Figure Q3 for which the parameters, referred to the stator, are:

stator resistance	0.04 Ω,
stator leakage reactance	0.4 Ω,
referred rotor resistance	0.055 Ω,
referred rotor leakage reactance	0.4 Ω,
magnetising resistance	20 Ω,
magnetising reactance	8 Ω.

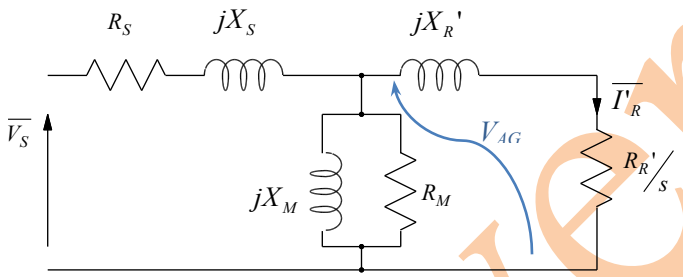


Figure Q3

The machine is rated to develop a torque of 500 Nm at a slip of 0.03. Calculate the following for operation at rated torque:

i) the rotor speed in rad/s [2]

First find synchronous speed and then apply slip equation

$$\omega_s = \frac{\omega_e}{p} = \frac{2\pi 50}{2} = 157.08 \quad rad/s$$

$$\omega_R = \omega_S\left(1-s\right) = 157.08 \times \left(1-0.03\right) = 152.36 \quad rad/s$$

[Accepted answers of 48.5π rad/s, did not accept answers in rpm.]

ii) the mechanical output power [3]

$$P_{Mech} = T\omega_R = 500 \times 152.36 = 76.2 \quad kW$$

[Occasional mistake was to divide the speed in rad/s by 2π. The proper units for angular velocity are rad/s and no conversion is needed.]

iii) the magnitude of the referred rotor current required to develop this power [5]

Take standard equation for mechanical power as function of rotor current and re-arrange. [3 marks for method; 2 for accurate answer]

Answers

$$P_{Mech} = \frac{3\, I_R^{'2}\, R_R^{'}}{\omega_R}\left(\frac{1-s}{s}\right)$$

$$I_R^{'} = \sqrt{\frac{P_{Mech}}{3\, R_R^{'}}\left(\frac{s}{1-s}\right)}$$

$$= \sqrt{\frac{76{,}200}{3\times 0.055}\left(\frac{0.03}{1-0.03}\right)} = 119.5 \quad A$$

[Most candidates answered this well with no clear pattern in mistakes.]

iv) the voltage appearing across the rotor branch (denoted V_{AG} in the equivalent circuit) in magnitude and angle form [5]

With the rotor current established in (iii), multiplication by the rotor branch impedance will yield the voltage. [3 marks for method; 2 for accurate answer]. Angle given with respect to rotor current.

$$Z'_R = \frac{R'_R}{s} + jX'_R = \frac{0.055}{0.03} + j0.4 \quad \Omega$$

$$V_{AG} = I'_R Z'_R = 119.5\times\left(\frac{0.055}{0.03} + j0.4\right) = 224.3\angle 12.3° \quad V$$

v) the magnetising current and stator current (in any convenient form) [5]

The rotor branch voltage established in (iv) also appears across the magnetising components and so currents through them can be found as two separate terms and summed to give magnetising current. Adding on the rotor current yields the stator current. [2 marks for method; 3 for accurate answers]. Angle given with respect to rotor current.

$$V_{AG} = 224.3\angle 12.3° \quad V$$

$$I_{RM} = \frac{V_{AG}}{R_M} = \frac{224.3\angle 12.3°}{20} = 11.21\angle 12.3° = 10.95 \; + \; j2.39 \quad A$$

$$I_{XM} = \frac{V_{AG}}{jX_M} = \frac{224.3\angle 12.3°}{j8} = 28.03\angle -77.7° = 5.97 \; - j27.39 \quad A$$

$$I_M = I_{RM} + I_{XM} = 16.93 \; - j25.00 \quad A$$

$$I_S = I_M + I'_R = 136.43 \; - j25.00 \quad A$$

[An occasional error was to fail to treat R_M and X_M as being in parallel with each other (and in parallel with the rotor branch, by for instance writing $I_M = \frac{V}{R_M + jX_M}$.

An alternative route which some candidates successfully applied was to use the current divider rule in reverse.]

vi) the magnitude of the required stator voltage [5]

Adding the stator voltage drop to the rotor branch voltage yields the required stator voltage (alternatively one could multiply the stator current by the total impedance of the machine). [2 marks for method; 3 for accurate answer]

Angle given with respect to rotor current.

$$Z_S = R_S + jX_S = 0.04 + j0.4 \quad \Omega$$

$$V_S = V_{AG} + I_S Z_S = 224.3\angle 12.3° + (136.43 \; - j25.00)(0.04 + j0.4)$$

$$= 234.5 + \text{j}101.4 \quad \text{V}$$

Answers

Which is a magnitude of 255.5 V

[A common alternative route to this answer was to calculate the total impedance of the machine and multiple by the stator current. It is also possible to apply the voltage divider equitation in reverse.]

vii) the efficiency at the rated torque [5]

Input power is found from stator voltage and current and output power was found in (ii). [2 marks for method and 3 for accurate answer]

$$P_I = 3\mathrm{Re}\left\{V_S I_S^*\right\} = 3\mathrm{Re}\left\{(234.5 + \mathrm{j}101.4)\times(136.43 + j25.00)\right\}$$
$$= 88.39 \quad kW$$
$$\eta = \frac{P_{Mech}}{P_I} = \frac{76.2}{88.39} = 86.2\%$$

[An occasional error was to neglect the factor of 3 arising from the three-phase system.]

Answers